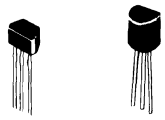
2SC536, 536K



2000B 2003A

NPN Planar
Silicon Transistors

Small Signal, General-Purpose Amp Applications

Ⓔ333H

Features

- The 2SC536 is classified into 2 types of SP, NP according to the case outline. The NP type is subclassified into 2 subtypes according to the breakdown voltage.
- Suitable for use in a wide range of applications from low frequency amplification to AM IF amplification.

Absolute Maximum Ratings at T_a=25°C

		2SC536SP	2SC536NP	2SC536KNP	unit
Collector to Base Voltage	V_{CBO}	40	40	55	V
Collector to Emitter Voltage	V_{CEO}	30	30	50	V
Emitter to Base Voltage	V_{EBO}	5	5	5	V
Collector Current	I_C	100	100	100	mA
Peak Collector Current	i_{cp}	300	300	300	mA
Collector Dissipation	P_C	200	400	400	mW
Junction Temperature	T_j	125	125	125	°C
Storage Temperature	T_{stg}	-40 to +125	-55 to +125	-55 to +125	°C

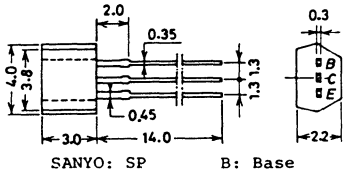
Electrical Characteristics at T_a=25°C

			min	typ	max	unit
Collector Cutoff Current	I_{CBO}	V_{CB}=35V, I_E=0			1.0	uA
Emitter Cutoff Current	I_{EBO}	V_{EB}=4V, I_C=0			1.0	uA
DC Current Gain	h_{FE}*	V_{CE}=6V, I_C=1mA	60*	320	960*	
Gain Bandwidth Product	f_T	V_{CE}=6V, I_C=1mA		100		MHz
Output Capacitance	C_{ob}	V_{CB}=6V, f=1MHz [2SC536SP]		3.0		pF
		[2SC536NP,KNP]		3.5		pF
Collector-Base Time Constant	$r_{bb'}C_C$	V_{CB}=6V, I_C=1mA, f=31.9MHz		250		ps
C-E Saturation Voltage	$V_{CE(sat)}$	I_C=50mA, I_B=5mA			0.5	V

*2SC536 is graded as follows by h_{FE} at 1A :

60	D	120	100	E	200	160	F	320	280	G	560	480	H	960

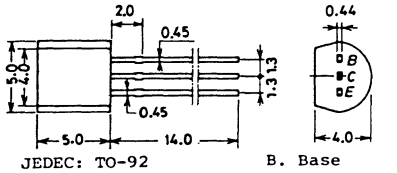
Case Outline 2000B (unit:mm)



SANYO: SP

B: Base
C: Collector
E: Emitter

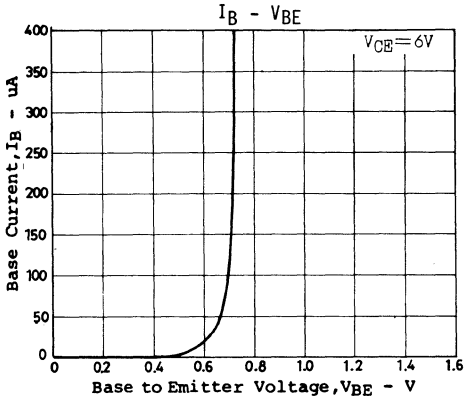
Case Outline 2003A (unit:mm)



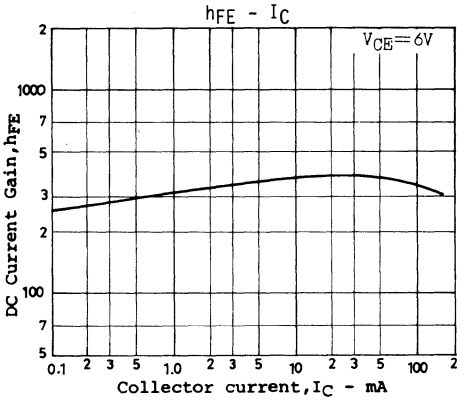
JEDEC: TO-92
EIAJ : SC-43
SANYO: NP

B. Base
C. Collector
E. Emitter

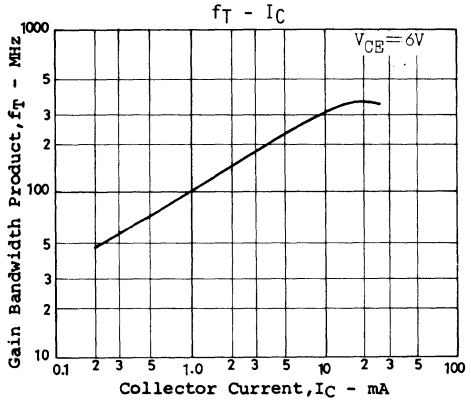
I_B - V_{BE}



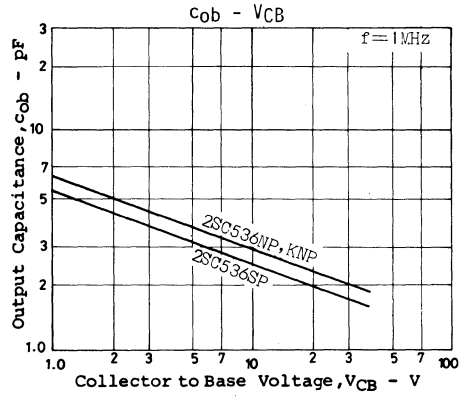
h_{FE} - I_C



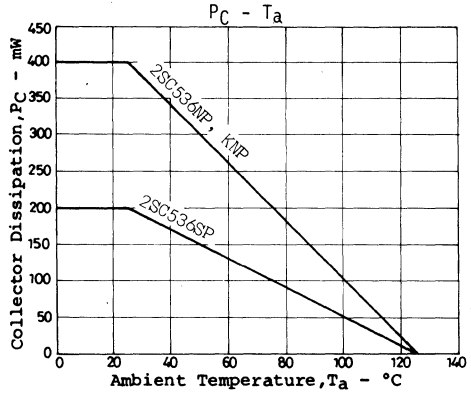
f_T - I_C



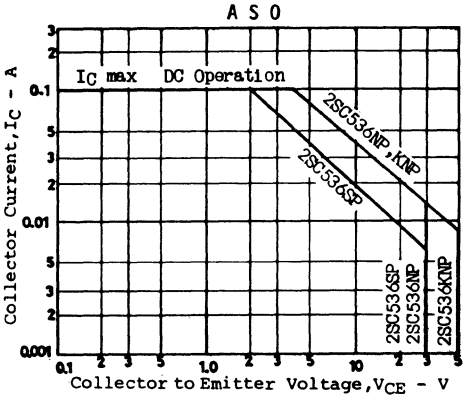
C_{ob} - V_{CB}



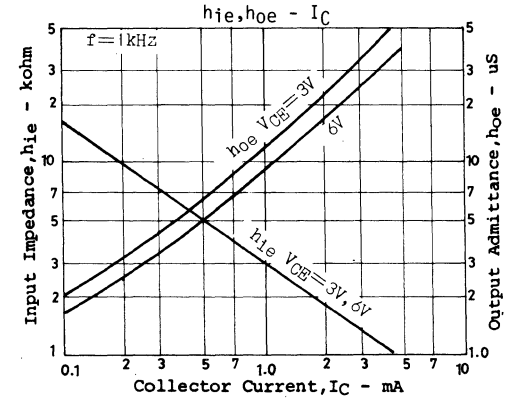
P_C - T_a



A S O



h_{ie}, h_{oe} - I_C



h_{re}, h_{fe} - I_C

